

JOANNE HOAR, M.Sc.

Software Developer • Technology Educator • AI Practitioner

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Hanna, Alberta, Canada • Remote

Summary

Canadian Software Developer and Technology Educator with a Master of Science in Computer Science (University of Calgary) and more than twenty years of experience building technical systems and teaching. Currently an active online instructor at Robertson College, where I teach and maintain curriculum across twelve programming and data subjects on Brightspace/D2L. I use AI tools daily across content development, code generation, CLI workflows, and process automation, and I teach students how to use AI responsibly, as a tool rather than a solution. I have developed and actively deliver curriculum on AI ethics, responsible use, and academic integrity as it applies to AI-generated content.

AI Expertise

Daily practice:

- Generative AI for content development: drafting course materials, case studies, scaffolded exercises, and code examples, always with accuracy review and editorial refinement
- Prompt engineering for structured output: system prompts, iterative refinement, chain-of-thought elicitation, output format and style control
- AI-assisted coding: **GitHub Copilot**, **Claude**, **ChatGPT**: for scaffolding, debugging, refactoring, and code review
- CLI and workflow automation: using AI to interface with **Azure**, **GitHub**, and shell tooling
- Multi-model practice: Claude, ChatGPT, GitHub Copilot, DeepSeek, Recraft AI, with different tools for different tasks

What I teach about AI:

- Responsible AI use as professional practice: practitioners are ultimately accountable for everything they produce, with or without AI assistance
- **Three-level AI assessment framework** (adapted from Perkins et al., 2024): from no-AI to AI-assisted task completion with mandatory human evaluation, applied directly to programming assignments
- Citation of AI-generated code: inline comment standards, README disclosure, terms of service compliance, with real code examples across seven scenarios
- Copyright, bias, hallucination detection, and institutional policy: applied to software contexts, not just essay writing
- AI tools as a first draft requiring editorial judgment, not a finished product

AI and assessment design (from Robertson's academic integrity training and the Academic Integrity Inter-Institutional Meeting, AIIIM 2026):

- "Authentic" learning and assessment: assignments grounded in real, local contexts, where generic AI output cannot succeed without genuine human judgment
- Scaffolded assignments with checkpoints: milestone grading and visible work in progress rather than a single final deliverable
- Developing professional judgment of AI output: students audit, correct, and debug flawed, biased, or hallucinated AI drafts
- Peer review and peer grading of AI-assisted work against concrete rubrics

Teaching & Course Development

Technology Instructor

Jan 2025 – Present

Robertson College – Online, Brightspace / D2L (Full Stack Development & Data Analytics programs)

- Maintain and deliver all course materials for **12 subjects**, with **20 course offerings** taught since January 2025; contribute to course revisions and create supplemental materials: sample code repositories, reference sheets, recorded walkthroughs, and open-source GitHub archives
- Teach students to use AI responsibly in software development; deliver academic integrity curriculum on AI citation, policy, and ethics; developed a dedicated slide deck actively used each term
- Completed Robertson's academic integrity training; participated in the Academic Integrity Inter-Institutional Meeting (AIIIM 2026)
- Served as **contracted Subject Matter Expert** for the Full Stack Web Developer program: wrote its sections of the cross-program Tech Programs FAQ for Robertson's internal Program Knowledge course, then built and presented slides on the program to about forty Robertson team members at an internal tech training session with live Q&A (March 2025)
- Invited back for future sessions and ongoing FSD SME consultation following the engagement
- Conduct 3+ synchronous live coding sessions per week; use real-time debugging as deliberate instructional modelling of professional problem-solving
- Apply Universal Design for Learning throughout: captions on recordings, structured documents, hardware-agnostic assignments
- Student satisfaction: **4.5 / 5.0** teaching career-changing adult learners; feedback highlights real-world context, approachability, and responsiveness

Brightspace/D2L • Angular • React • ASP.NET Core • C# • Java • Git • Azure • SQL • Python

Instructor, Intermediate C++ Programming

2005 – 2008

University of Calgary, Department of Continuing Education – Adult Learners

- Designed and delivered complete curriculum for adult learners: outcomes, lectures, sample code, and assessments from the ground up

Research / Teaching Assistant

2001 – 2004

VLab / AI Research Hub, University of Calgary

- TA and lab instructor: *Introduction to Programming, Computing for Non-Majors, and Operating Systems*
- Computing for Non-Majors required translating technical concepts for a completely non-technical audience

Industry Experience

Senior Software Developer

2022 – 2025

bitHeads Inc. – brainCloud BaaS SDK

- Authored extensive developer-facing documentation, onboarding guides, and sample projects; hired in part for educational background
- Provided one-on-one client instruction across seven UI frameworks and twelve platforms

Co-founder / Lead Developer

2005 – 2022

Silicon Hanna Inc. (self-employed; founded as SuRJE Software Solutions Inc.)

- 17-year software development and consulting practice spanning mobile, web, AR, and games
- Early AI/ML research roots: M.Sc. in agent-based systems, swarm intelligence, and evolutionary computation, the academic precursors to today's machine learning landscape

- Published 11 mobile titles (iOS and Android); explored novel SDKs and produced learning materials throughout

Tools Programmer

2008 – 2011

Fekete Associates Inc. (S&P Global)

- Designed and delivered a voluntary **internal graphics training course** for colleagues from beginners to senior developers; built multi-level materials with a hands-on capstone

Intermediate Software Engineer

2005 – 2008

BJ Pipeline Inspection Services Inc. (Baker Hughes)

- Implemented **neural network algorithms** for automated anomaly detection in pipeline inspection data: applied machine learning in industry well before the current AI wave

Education

- M.Sc. Computer Science** – University of Calgary 2004
Thesis: Agent-based Development of Natural Transportation Networks
Graduate coursework: Human-Computer Interaction, Artificial Intelligence, Computer Graphics
- B.Sc. Computer Science** – University of Calgary 2001

Honours & Awards

- Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies 2005
- Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada 2005
- Graduate Student Scholarship, Alberta Learning 2004
- Best Student Paper Award (Co-author), Australian Conference on Artificial Life (ACAL) 2003
- CPSC Departmental Research Award, University of Calgary 2003

Selected Publications *(published under maiden name Penner)*

- Penner, J.**, Hoar, R., Jacob, C. "Bacterial Chemotaxis in Silico," *Australian Conference on Artificial Life (ACAL)*, 2003 *Best Student Paper*
- Hoar, R., **Penner, J.**, Jacob, C. "Transcription and Evolution of a Virtual Bacteria Culture," *IEEE Congress on Evolutionary Computation (CEC)*, 2003
- Hoar, R., **Penner, J.** "The Application of Artificial Intelligence to Transportation System Design," *XRDS: Crossroads, ACM*, 2003
- Penner, J.**, Hoar, R., Jacob, C. "Swarm-Based Traffic Simulation with Evolutionary Traffic Light Adaptation," *IASTED ASM*, 2002
- Hoar, R., **Penner, J.**, Jacob, C. "Evolutionary Swarm Traffic: If Ant Roads Had Traffic Lights," *IEEE CEC*, 2002

Technical Skills

AI Tools: ChatGPT, Claude, GitHub Copilot, DeepSeek, Recraft AI, used responsibly in curriculum, content, and coding workflows

LMS: Brightspace / D2L (daily), GitHub (open-source course repos)

Languages: C++, C#, JavaScript, TypeScript, Java, Python, PHP, SQL

Frameworks: Angular, React, ASP.NET Core, Node.js, .NET

Content & Authoring: Markdown, $\text{\LaTeX}$, HTML/CSS, open-source tooling

Human Languages: English (native), French (intermediate), Spanish (rudimentary)

Work Samples

🔗 github.com/xcodegirl/career (xcodegirl-work-samples.md)

Sample lesson plans and slide decks available upon request.